
MODEL SELECTION AND VALIDATION REPORT (general_tric model)

OBSERVATIONS:3382

SANITY: OK - all models in the selection process passed sanity checks.

SELECTED RANDOM TERM: spatiotemporal_random_fields_ar1

MODEL SUMMARY (general_tric model)

Spatiotemporal model fit by REML ['sdmTMB']

Formula: fixed_formula

Time column: year

Family: gaussian(link = 'identity')

Conditional model:

	coef.est	coef.se
(Intercept)	0.05	0.54
sbt_mean	0.26	0.06
sbt_sd	-0.03	0.03
sst_sd	-0.04	0.03
sic_mean	0.03	0.05
log_chla_mean	0.01	0.02
depth	0.03	0.03

Dispersion parameter: 0.77

Spatiotemporal AR1 correlation (rho): 0.97

Matérn range: 501.39

Spatiotemporal marginal AR1 SD: 1.17

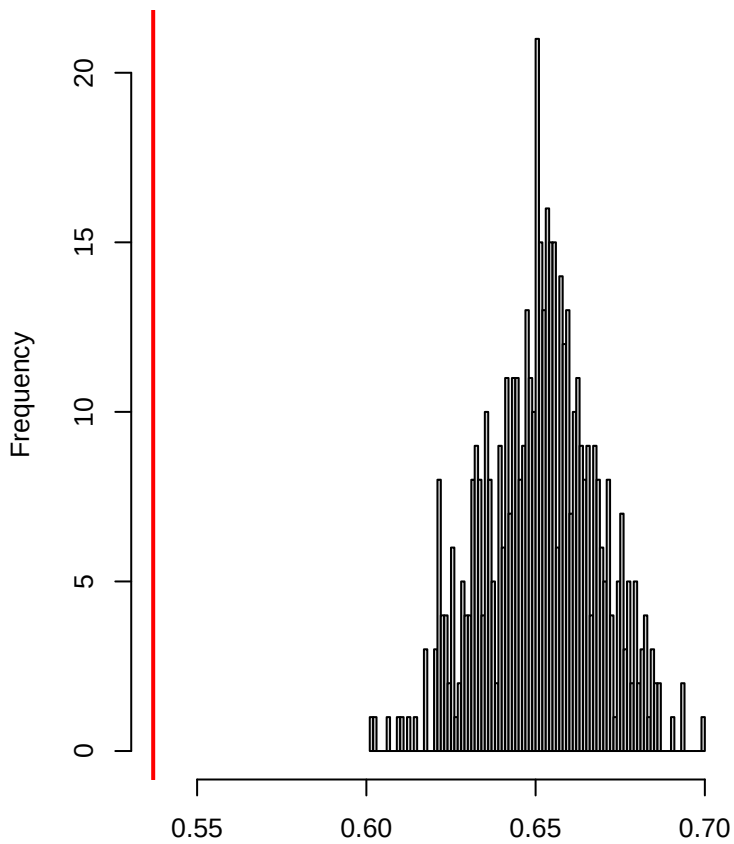
REML criterion at convergence: 4249.157

See ?tidy.sdmTMB to extract these values as a data frame.

SELECTION STATISTICS

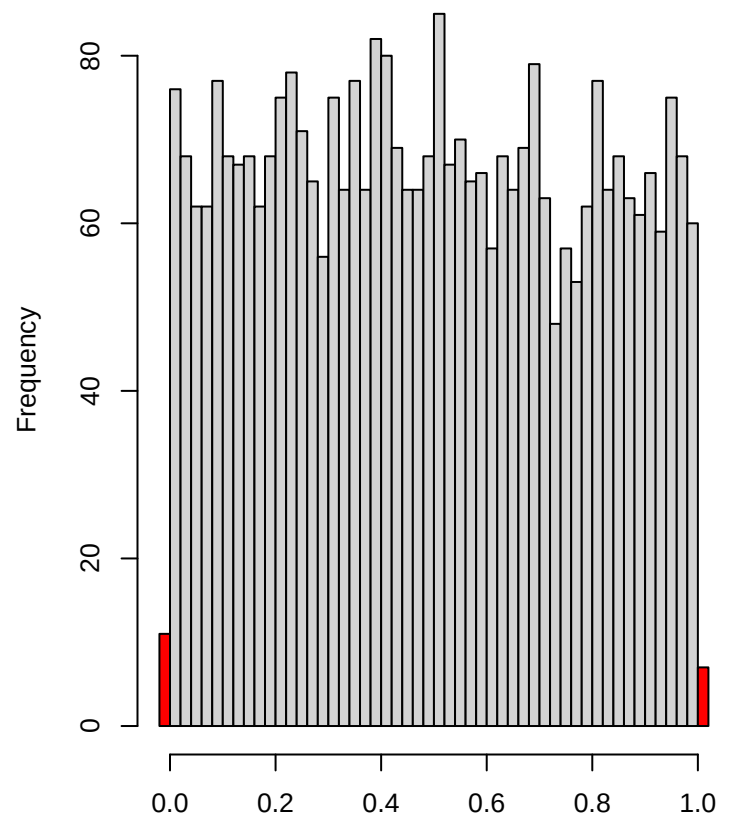
	df	AIC	delta_AIC
spatiotemporal_random_fields_ar1	11	8520.314	3.424953
spatiotemporal_random_fields_ar1_with_anisotropy	13	8523.739	375.483655
spatiotemporal_random_fields_iid_with_anisotropy	12	8899.223	1.791457
spatiotemporal_random_fields_iid	10	8901.014	109.353856
spatial_random_fields	10	9010.368	3.869956
spatial_random_fields_with_anisotropy	12	9014.238	558.551948
error_only	8	9572.790	NA

DHARMa nonparametric dispersion test via sd of residuals fitted vs. simulated



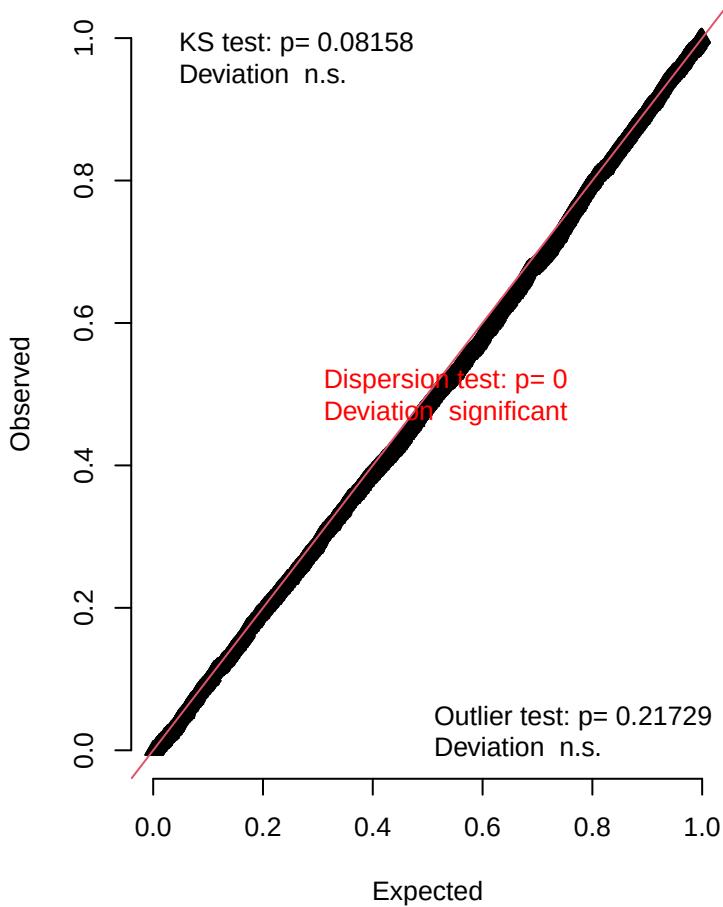
Simulated values, red line = fitted model. p-value (two.sided)

Outlier test n.s.

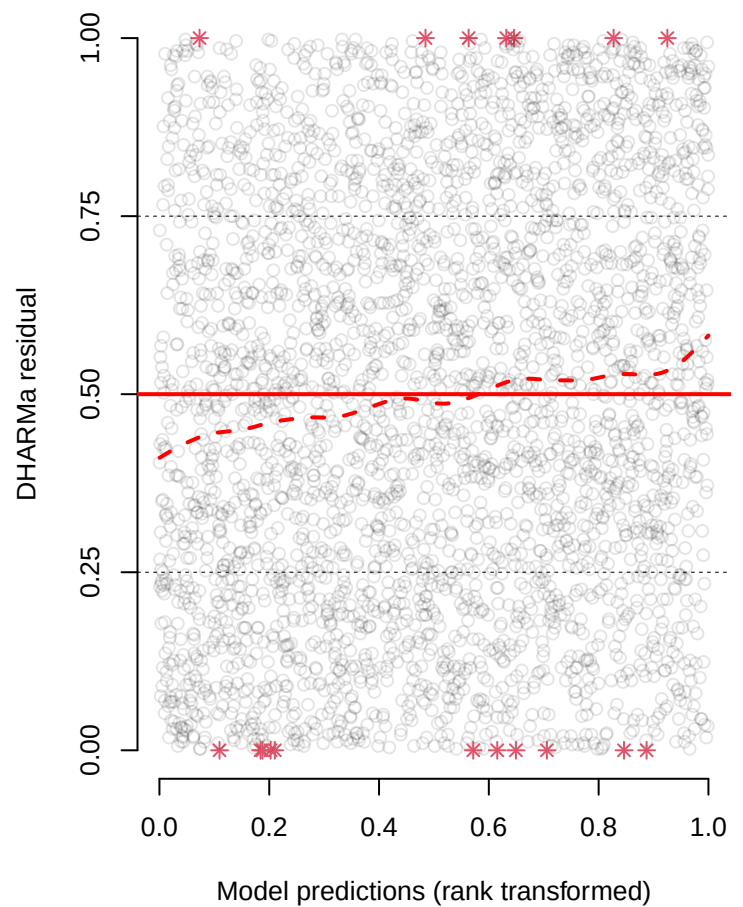


Residuals (outliers are marked red)

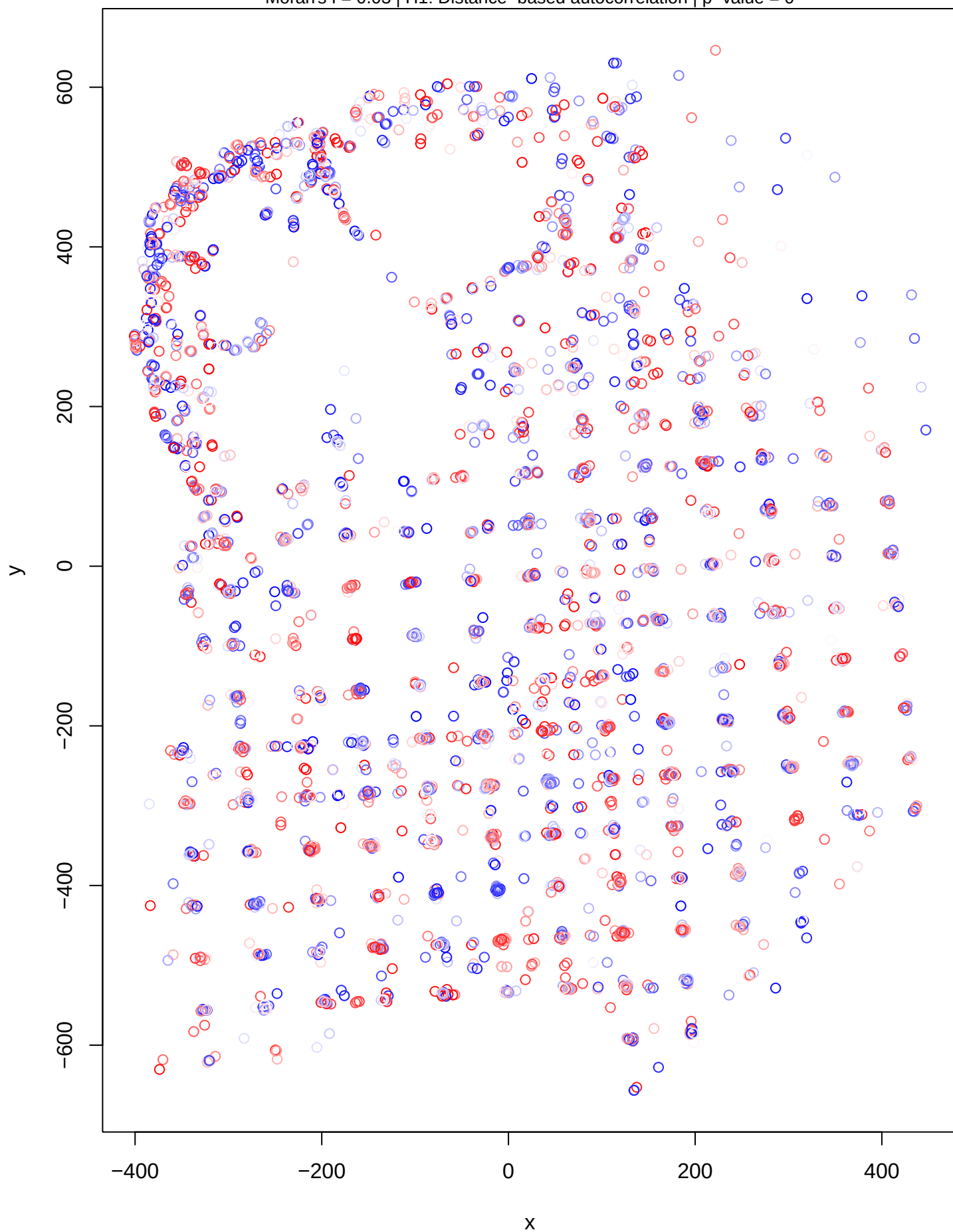
QQ plot residuals



DHARMa residual vs. predicted

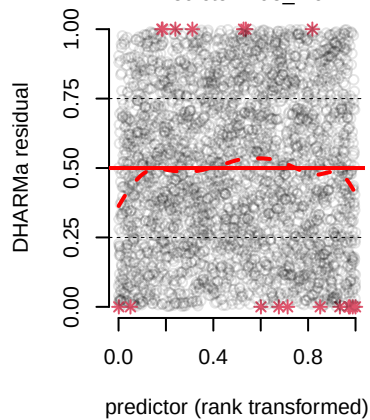


DHARMa Moran's I test for distance-based autocorrelation
Moran's I = 0.03 | H1: Distance-based autocorrelation | p-value = 0



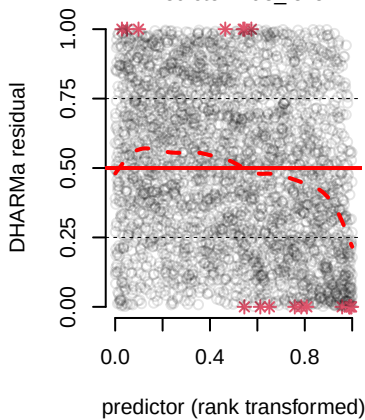
DHARMa residual Residual vs. pred

Predictor: kde_fric



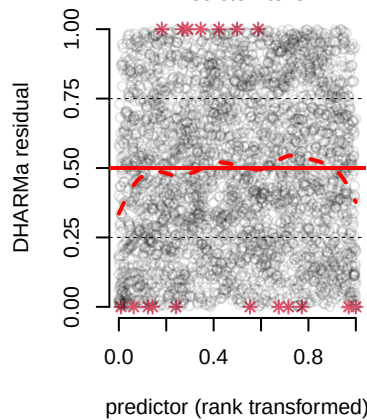
DHARMa residual Residual vs. pred

Predictor: kde_feve



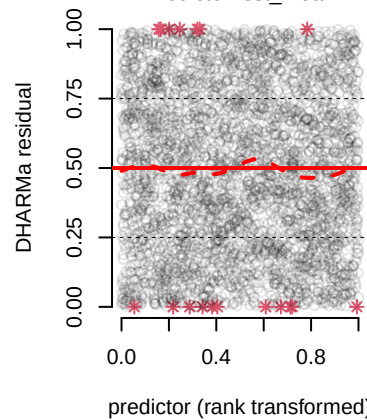
DHARMa residual Residual vs. pred

Predictor: teve



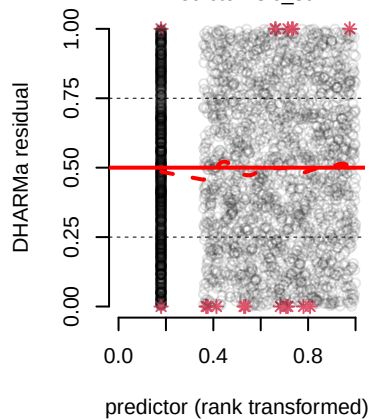
DHARMa residual Residual vs. pred

Predictor: sst_mean



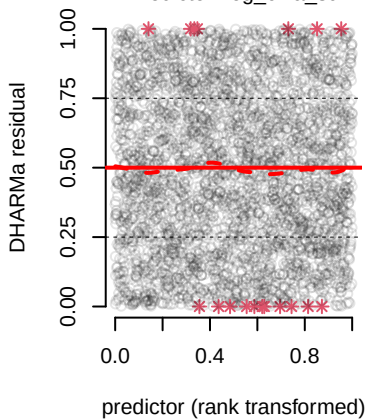
DHARMa residual Residual vs. pred

Predictor: sic_sd



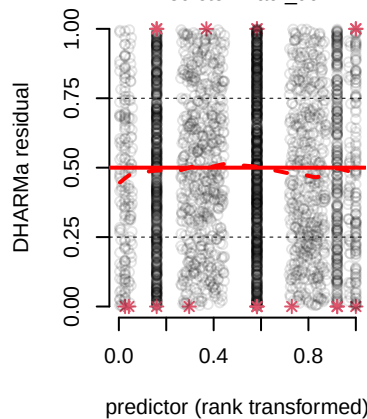
DHARMa residual Residual vs. pred

Predictor: log_chla_sd



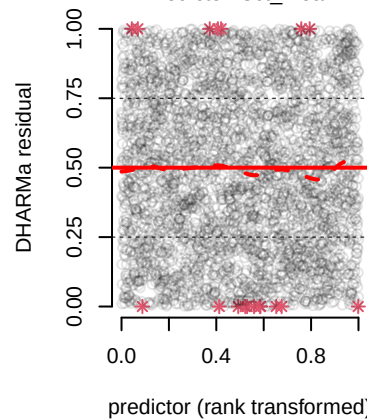
DHARMa residual Residual vs. pred

Predictor: haul_dur



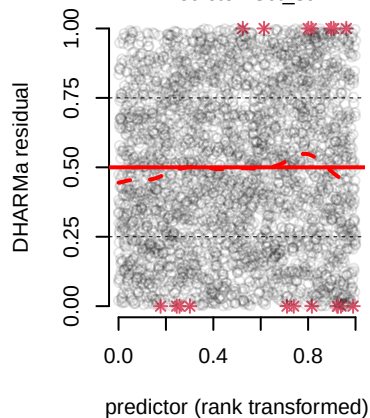
DHARMa residual Residual vs. pred

Predictor: sbt_mean



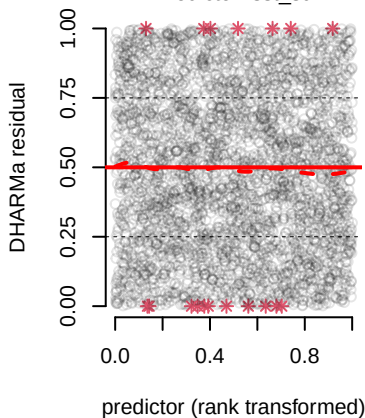
DHARMa residual Residual vs. pred

Predictor: sbt_sd



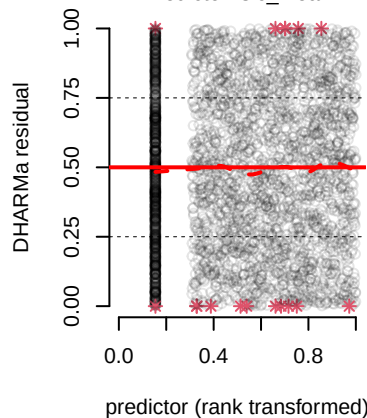
DHARMa residual Residual vs. pred

Predictor: sst_sd



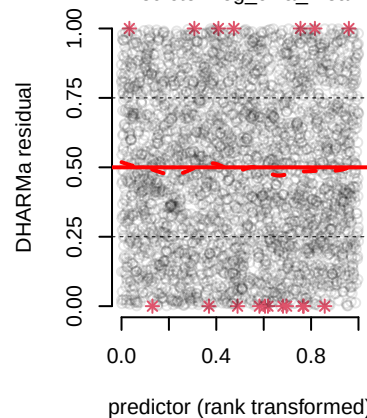
DHARMa residual Residual vs. pred

Predictor: sic_mean



DHARMa residual Residual vs. pred

Predictor: log_chla_mean



DHARMa residual Residual vs. pred

Predictor: depth

